PCB COST DRIVERS

What drives the cost of a circuit board?

And how do those factors in turn impact the sustainability?

Whether your role is managerial, commercial or technical there is a good chance that you have an interest in ensuring that the printed circuit boards you are designing, purchasing and/or assembling are as efficient as they can be in terms of achieving the lowest total cost, sustainably, without compromising functionality or reliability of the end product.

The majority of the cost – 80-90%- is built into the product in the early stages of the design phase –before either the PCB supplier or EMS company has sight of the design. We have also found that many of the cost drivers directly impact all aspects in producing and delivering a more sustainable product. In our Sustainability Impact meter you can see the effects for each cost driver.

NCAB has divided the cost drivers into two categories, the 'hard costs' and the 'soft costs.' The hard cost drivers are covering aspects that relate to the physical printed circuit board itself. Examples being copper weights and build complexity. In these cases the impacts on sustainability are more tangible. The soft cost drivers are those elements that unless considered can lead to delays in engineering, lost time and mis-specified demands that only serve to either drive cost upwards or fail to identify what is needed to achieve the desired quality or long-term reliability – this is not sustainable PCB production in our eyes. Many times these factors are a little less quantifiable, but can certainly have just as much of an impact on cost and sustainability.

HARD COST DRIVERS

1. **PCB size**
2. **Layer count**
3. **Build / Complexity**
4. **Via treatments**
5. **Material / panel utilization**
6. **Track and gap**
7. **Hole (size & quantity)**
8. **Additional drilled/routed features**
9. **Hard gold / Thick gold**
10. **Impedance**
11. **Excessive tolerances**
12. **Copper foil weights**
13. **Soldermask, legend/silkscreen, carbon print**
14. **Surface finishes**
15. **Materials**

SOFT COST DRIVERS

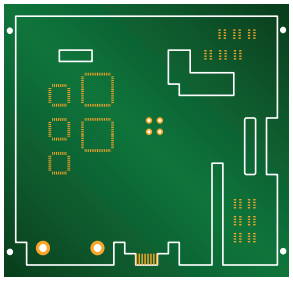
16. **Under specification**
17. **Over specification**
18. **Reliability**
19. **Lead times**
20. **Transportation costs**
21. **Lack of knowledge, communication, willingness**

Hard cost drivers

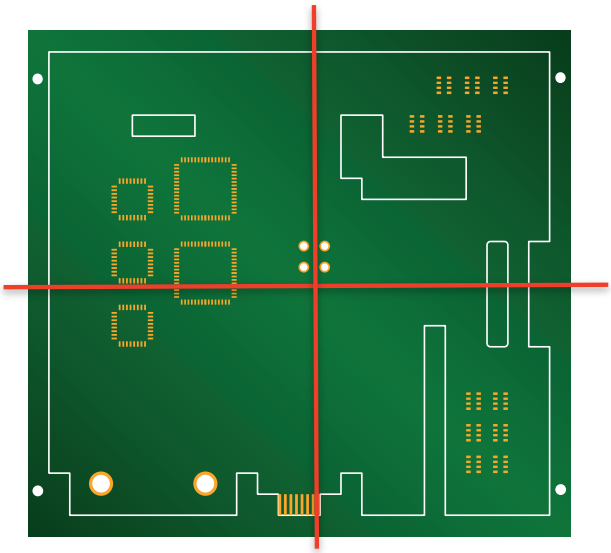
1 The PCB size

This is one of the very simplest drivers to understand, just as with real estate - the larger it is, the more it costs. And aside from simply the use of more material, we need to consider:

- Should a defect occur in a smaller circuit, the remainder of the panel can still be acceptable for production
- When a defect occurs in a larger circuit a larger portion of raw material is discarded
- Environmental impact of material demand, less material = lower environmental impact



SIZE 150*150MM
Price x1
(base price)



SIZE 300*300MM
Price x4+

Sustainability Impact

IF WE ARE ABLE TO REDUCE THE SIZE

- Less environmental impact producing the materials (water, heat, electricity)
- Reduction in raw materials (copper, fiberglass)
- Less waste if a defect occurs and board must be scrapped
- Reduced packaging / Co2 due to smaller profile (freight)

2 Layer count

Layer count is another driver that's easy to understand. Much like the physical size of the PCB, more layers means more material required for the build. Apart from the raw material demand, there is also additional labor required to process each of the additional layers.

SPECIFIC REASONS FOR LAYER COUNT DRIVING COST

- Each layer will have to go through the same various processes
- Cost for each of the inner layers / dry film
- Increased material cost and processing cost
- Developing / Etching / AOI inspection for each inner layer
- Labor costs
- Black / brown oxide for each inner layer
- Cost for each piece of pre-preg used between layers
- Pressing / bonding cycles necessary

THE FIGURES BELOW ARE AVERAGE COST ADDERS BASED UPON A CROSS SECTION OF FACTORIES. AS SUCH THESE FIGURES SHALL ONLY BE CONSIDERED AS A GUIDE OR POINT OF REFERENCE.

1	2 layer	**+45%**
2	4 layer	**+40–45%**
4	6 layer	**+35–40%**
6	8 layer	**+30–35%**
8	10 layer	**+25–30%**
10	12 layer	**+25–30%**
12	14 layer	**+15–20%**

Sustainability Impact

IF WE CAN REDUCE THE NUMBER OF LAYERS

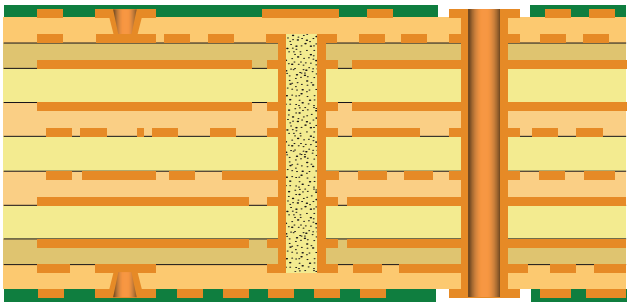
- Less environmental impact producing the materials (water, heat, electricity)
- Reduced labor required to process each of the additional layers
- Reduced materials required for each of the inner layers
- Reduced packaging / Co2 due to less freight weight
- Reduced production time
- Reduced labor

3 Build / Complexity

Much like layer count, complexity drives cost due to the additional labor for processing required. Some complex HDI builds can 'loop' multiple processes like lamination, drilling and plating several times making their production timely and also relatively expensive when compared to a simple multilayer of the same size and layer count.

Another complex requirement that drives cost here is laser drilling for microvias which is inefficient when compared to mechanically drilling multiple production panels at one time. Unfortunately as we have seen, complex builds are almost always necessary to achieve a product's application or footprint but caution should be used to ensure these features are applied in a cost-effective manner.

10L PCB	Base price
Layer of microvia **(1+8+1)**	**+40–70%**
Buried via and microvia **(1+8b+1)**	**+80–120%**
Buried via and microvia with copper fill	**+120–140%**



Both PCB size and complexity multiply the sustainability impact. more layers and complexity of build multiplied by the PCB size directly impact the carbon footprint and environmental impact by increasing the materials / chemicals and transportation weight.

Sustainability Impact

IF WE ARE ABLE TO DESIGN A LESS COMPLEX BOARD

- Increased efficiency with ability to mechanically drill multiple panels at once without microvia
- Reduced cycles/processes (lamination, drilling, plating)
- Reduced production time
- Reduced labor
- Less materials/chemicals

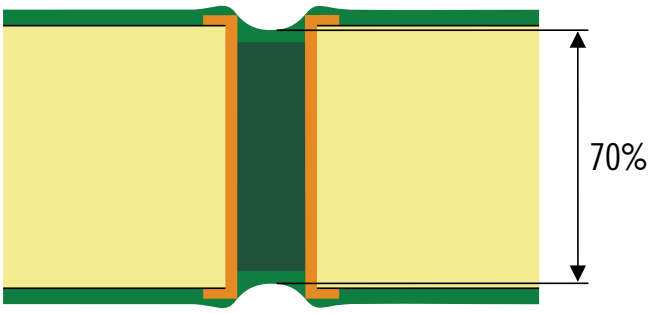
4 Via treatments

When designs become more complex we often have requirements for via in pad or via treatment to ease assembly, we refer to IPC 4761, to avoid via related problems in the future. With multiple solutions offered by this specification, each has a different cost driver. Knowing what you really need for assembly can ensure the most cost-effective solution is utilized.

The figures below are average based upon a cross section of factories. As such these figures shall only be considered as a guide or point of reference.

Via treatment cost impact

Soldermask covered (Tented)	**+0%**
Soldermask plug	**+1–5%**
Type VI	**+5–10%**
Type VII Nonconductive resin	**+7–12%**
Type VII Conductive resin	**20+%, subject to material MOV**



Sustainability Impact

IF WE ARE ABLE TO REDUCE OR ELIMINATE VIA TREATMENTS

- Less chemicals
- Elimination of type VII, less epoxy resin and plating process
- Reduced production time
- Reduced labor

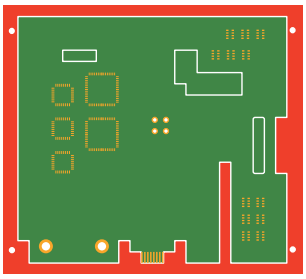
Via treatments bring into the build additional chemicals which impact sustainability. Type 7 being the largest contributor due to the addition of epoxy resin and additional plating processes.

5 Material / panel utilization

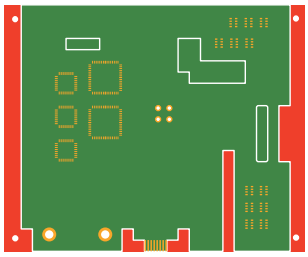
EXAMPLE 1

This should not be confused with our first cost driver, the size of the PCB. However, both are cost drivers. Utilization is the percentage of the working panel allocated to the specific array. Pricing is based on the entire working panel. If you have a PCB that takes up 2/3 of a panel, there is not enough space left on the panel to fit a second board but you still pay for all the wasted material. In addition, the material waste is not sustainable for cost or the environment.

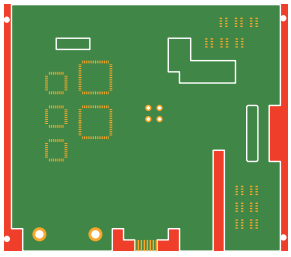
Consider the 3 panel options below. **RED** is the carrier rail. It has the same composition as the actual PCB but ends up in the recycling bin after the PCB is depaneled. Option 1 is a very common request however it comes at a premium over option 3 which has a higher panel utilization and also a smaller material demand due to the reduced number of smaller rails.



PANELIZATION 1



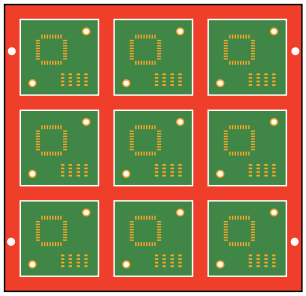
PANELIZATION 2



PANELIZATION 3

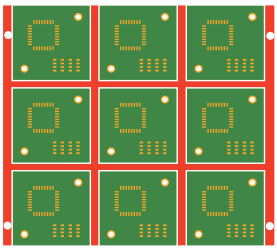
EXAMPLE 2

Here is another example of material utilization and demand. In the images below we can see the large waste strips (or wasted area in, panelization 1. In panelization 2 and 3 we have removed these waste areas. The smaller array allows more boards per panel and helps keep a high material utilization and also a low material demand. In these examples panelization 3 is the lowest cost option.



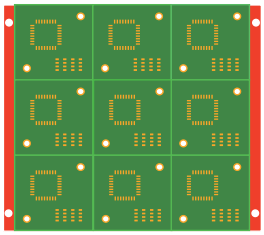
PANELIZATION 1

Panel option 1 = 1 panel of 9 circuits,
40mm x 40mm, 10mm carrier rail on 4x sides.
Route gap 2.4mm between all circuits
Total Panel size = 304.15 mm2



PANELIZATION 2

Panel option 2 = 5mm carrier rail on 2x sides.
Route gap 2.4mm between all circuits,
Total Panel size = 215.54mm2
Material reduction = 29%



PANELIZATION 3

Panel option 3 = 5mm carrier rail on
2x sides. Score lines between all circuits.
Total Panel size = 156.00mm2
Material reduction = 49%

Sustainability Impact

IF WE MAKE BEST USE OF SPACE ON THE PANEL

- Less environmental impact producing the materials (water, heat, electricity)
- Reduction in raw materials (copper, fiberglass)
- Less waste if a defect occurs and board must be scrapped
- Less scrap
- Reduced packaging / Co2 due to smaller profile (freight)

Less material = lower sustainability impact. low material utilization = high material waste. This is one of the largest opportunities for improvement on a global level but requires open communication between the designer and PCB fabricator.

6 Track and gap

HEADLINE 1

Smaller lines and smaller spaces drive cost because the ability to achieve finer features is dependent upon the capability of the factory. Every factory has different capabilities which will impact cost. With a smaller track and gap comes the risk of reduced yields which is even more of a factor when we consider the density and length of fine tracks and gaps.

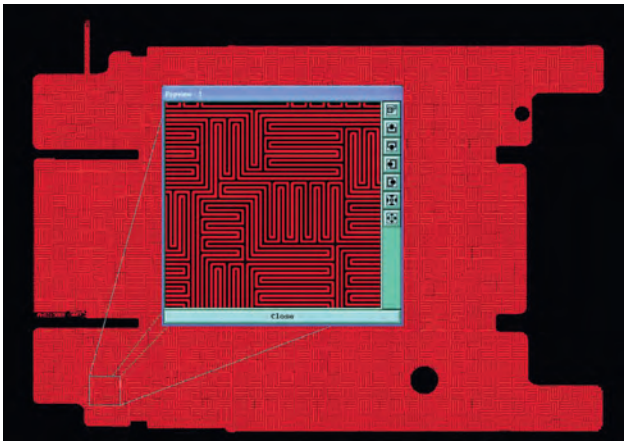
HOW DOES THIS INFLUENCE THE DESIGN AND COST?

- Length of fine line tracks
- Copper weight / base foil
- Exposure units
- Etching methods
- Handling methods
- Type of clean room
- Additive processing vs traditional imaging

HEADLINE 2

It is difficult to give exact figures how the track / gap affects the price. The reason for this is that while the standard is 5/5 mil track & gap, the cost is also driven by what yield is expected and this can depend on how many meters of fine line tracks are running in parallel.

The PCB (as shown) has several hundreds of meters of tracks in one single loop – the slightest issue with imaging / etching and the entire PCB is scrap.



Fine features should be used sparingly, if you don't need an entire layer to be fine featured only have as much miniaturization as needed.

Sustainability Impact

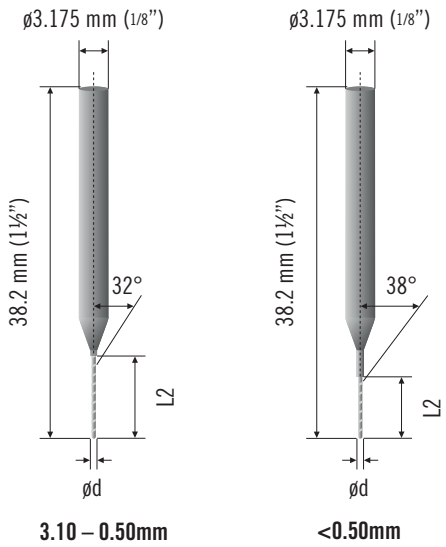
IF WE ARE ABLE DECREASE THE AMOUNT OF SMALLER/FINER TRACES AND SPACES

- It will increase yields and the amount of factories that can achieve production reliably.
- It will also minimize the risk of failure in production, waste and lead time.

7 Hole (size & quantity)

DRILLING OF SMALL HOLES IS A COST DRIVER

Smaller finished hole sizes need smaller drill bits. Smaller drill bits have shorter flutes which means less panels can be drilled at a time reducing the efficiency of drilling. Additionally, the smallest common mechanical drill available for low cost volume manufacturing is .2mm. When smaller holes are required for mass production we will need to build these parts in higher cost regions or a prototype facility.



This graphic shows two mechanical drill bits, the smaller having a shorted flute which is the part of the bit that does the actually drilling.

Sustainability Impact

IF WE ARE ABLE TO REDUCE THE NUMBER OF HOLES AND INCREASE THE SIZE

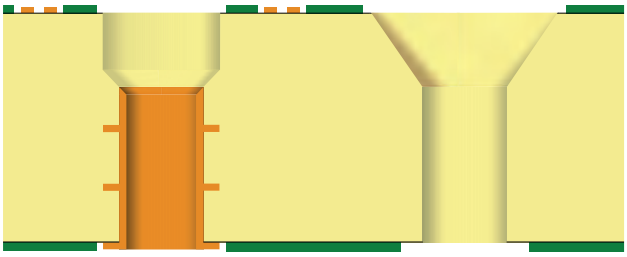
- Increased efficiency with ability to drill more panels at once with larger, more robust drill bit

8 Additional drilled/routed features

All of these add to cost as they are additional processes and require non-standard drill tooling. The cost adder is based on the actual size of the feature, the number of hits and the cost of the tool. These directly attribute to time on a router and labor for supporting the process.

If the feature is an uncommon size MOV for the tool and additional lead time to procure or purchase the tool may apply.

- Counter bores
- Countersinks
- Backdrilling
- Controlled depth milling
- Double sided routing processes.



Sustainability Impact

IF WE ARE ABLE TO REDUCE THE AMOUNT OF DRILLED/ROUTED FEATURES, OR THE SIZE

- Scoring vs. Routing reduces the excess fiberglass removal requiring waste hauling
- Decreased labor - washing post route to remove dust

9 Hard gold / Thick gold

Gold remains expensive and hard gold finishing can be the most expensive surface treatment adder depending upon the total surface area to be plated. In this process we are plating gold with a hardener over nickel. A small spot compared to a large area can have very different costs. Hard gold plating is a non-standard process so any amount of gold is also affected by a labor cost for the additional process.

EXAMPLES OF COMMON COST ADDERS FOR HARD/THICK GOLD:

- Extra dry film process
- Additional space necessary for tracks to connect necessary areas
- Less panel utilization with boards with slot-in contacts

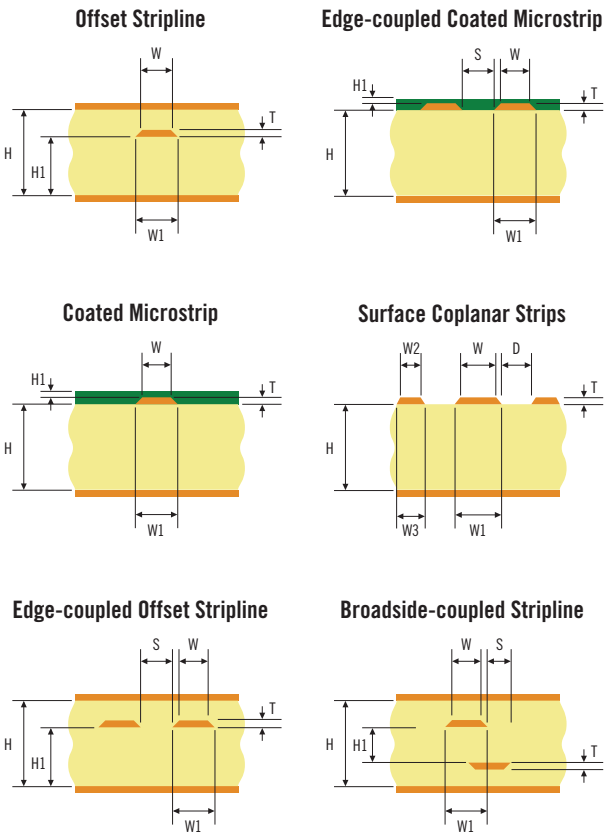
Sustainability Impact

IF WE CAN DECREASE THE AMOUNT OF HARD AND/OR THICK GOLD

- Decrease film processes, (electricity, time and labor)
- Decrease space/materials

10 Impedance

- Controlled impedance is a cost driver because it demands very specific controls on line width and spacing along with the dielectric spacing. The production panel must have extra space available for the testing coupons and the actual testing of the coupons via Time Domain Reflectometer test (TDR) is a non-standard process requiring labor.
- This graphic illustrates some common track structures for impedance variables along with the variable type.



Sustainability Impact

IF WE CAN REDUCE IMPEDANCE NEEDED

- Reduce panel space needed / materials
- Reduced processes / labor

11 Excessive tolerances

- Annular rings below 150µm
- Outline dimensions tighter than +/- 0.10mm
- Tracks smaller than 100µm
- Gaps below 100µm
- All hole sizes +/- 0.05mm
- Aspect ratio above 1:8
- Impedance tighter than +/-10%
- Line width tolerance tighter than IPC standard

Tight tolerances can result in a design where perhaps only a single factory can support. This is never a good situation. This also turns a basic/standard PCB into an unnecessarily complex one.

If such tolerances are not critical to the fit/form/function of the PCB, then often it can be that these overly strict tolerances are unnecessary.

Sustainability Impact

IF WE ARE ABLE TO ELIMINATE UNNEEDED EXCESSIVE TOLERANCES

- Minimize production batch so post fabrication there's sufficient yield
- Increased factory options able to meet requirements, increasing competitive pricing

Large sustainability impact here as tolerances require a larger batch put into production so that post fabrication we have a sufficient yeild to meet the customers demand.

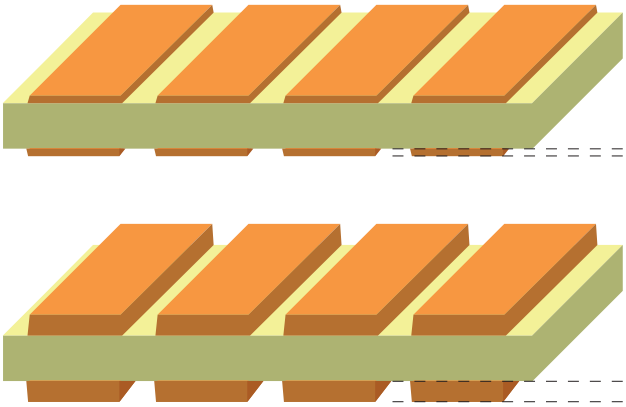
12 Copper foil weights

As with most things in life, copper costs money, so the thicker the copper, the higher the cost. Copper is also rather heavy so thicker copper causes the PCBs to be heavier, resulting in higher freight charges.

Thick copper on inner layers requires more prepreg during pressing to fill up all cavities between copper features.

Track and gap is more difficult to control on heavy copper – as the copper weight increases so does the track and gap requirement. It may also need additional soldermask application to coat the knee of the tracks.

Factories secure UL approvals by copper weight and heavier-weight copper becomes less and less common for a factory to have an existing UL approval. Because of this as copper weight increases the number of capable factories decreases.



Ref copper weights IPC-6012!

Sustainability Impact

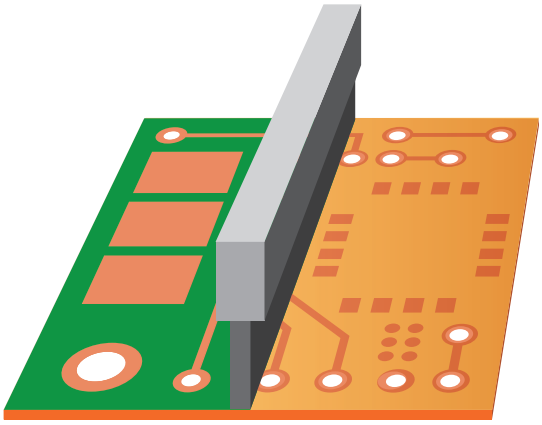
IF WE ARE ABLE TO REDUCE THE AMOUNT OF COPPER

- Reduced weight and in turn reduced shipping costs
- Reduced weight and in turn reduced carbon footprint from shipping
- Reduced copper weight requires more less time in etching which means less chemicals are required to produce each layer

From a sustainability standpoint heavier copper requires more time in etching which means more chemicals are required to produce each layer. It also means the carbon footprint is larger from transportations due to the weight of the finished PCB's.

13 Soldermask, legend/silkscreen, carbon print

- Each one is a separate screen print / cure operation
- Cost can be in the region of 2-5% extra dependent upon technology
- Specific soldermask types and colors (high chemical resistance, or blue soldermask for example) can also add more to the price due to material cost, complexity of application or available factory capability
- Thick soldermask requirements or heavy copper may require multiple coats increasing labor and throughput



Sustainability Impact

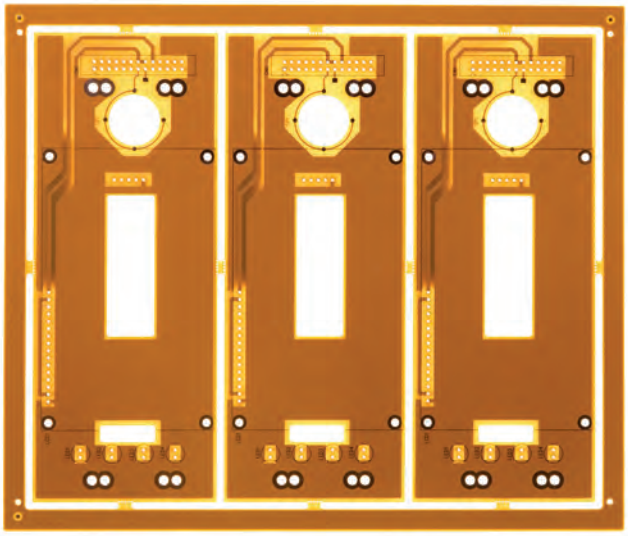
IF WE ARE ABLE TO ELIMINATE OR REDUCE SPECIALIZED SOLDER MASK REQUIREMENT AND/OR LEGEND SILKSCREEN

- It will increase the number of factories that can support, which will drive the cost down
- Thinner soldermask decreases weight and transportation carbon footprint
- Decreased number of coats reduces chemical usage

14 Surface finishes

As the adder for surface finish is a constant, it has a higher percentage impact on lower technology boards compared to higher technology boards. The actual cost adder is dependent on the surface finish, the percentage of surface area requiring the finish as well as whether the finish is an in-house process or one that's outsourced.

Type	2-Layer	12-Layer
OSP	0%	0%
HASL	0%	0%
LF HASL	0 - 4%	0 - 1%
Imm Ag	18%	4%
Imm Sn	18%	4%
ENIG	28%	6%
ENEPIG	By quote	By quote



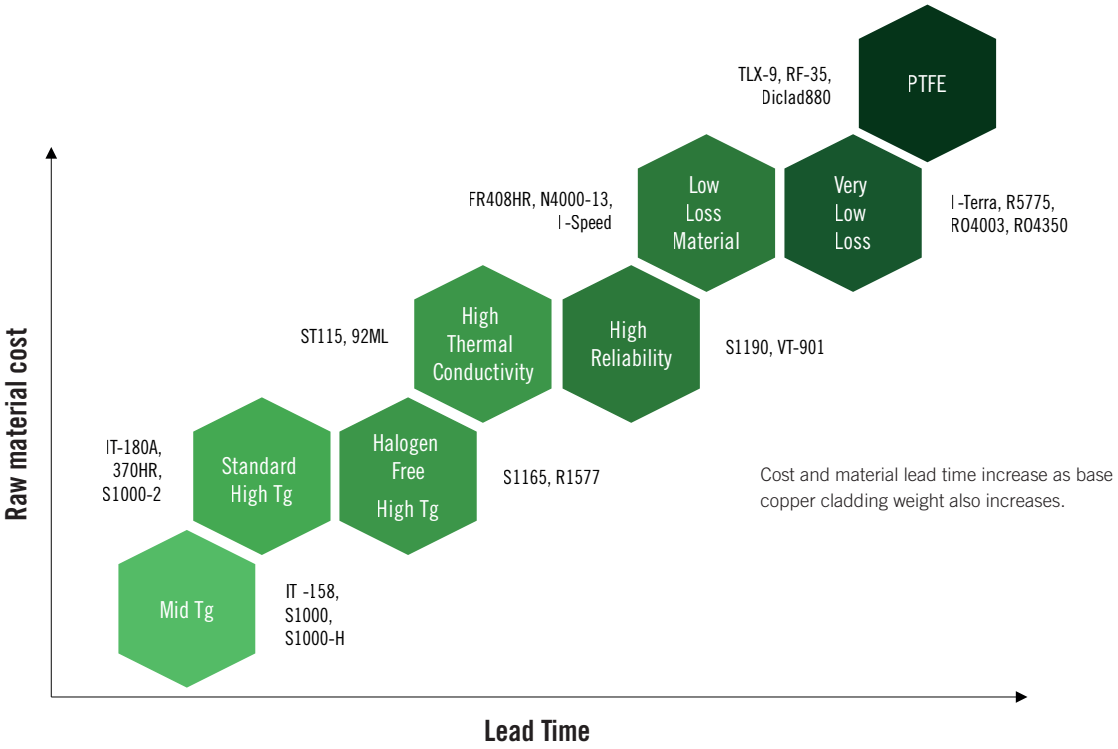
Sustainability Impact

IF WE ARE ABLE TO USE MORE COMMON SURFACE FINISHES

- Increased the available factories that can support materials on hand
- Lighter weight surface finishes weight and transportation carbon footprint

15 Materials

Base material is a cost driver because some materials cost more than others. Also because some materials are harder to process than others and lastly because certain materials may only have select factories with UL approval for their use.



Sustainability Impact

IF WE ARE ABLE TO USE MORE COMMON BASE MATERIALS

- Increased the available factories that can support materials on hand
- Surface finishes and copper weight offer potential to reduce transportation cost and carbon footprint.

Soft cost drivers

16 Under specification

NCAB Group often experiences examples of under-specification (lack of vital information) resulting in time consuming review / investigation. This causes costly delays, confusion, and prevents acceptance expectations being fully understood.

SOME TYPICAL EXAMPLES OF MISSING INFORMATION

- Contour detail
- Lack of detail regarding PTH vs. NPTH designations
- Surface finish not specified
- Copper thickness
- Base material information not provided
- Soldermask color
- Thickness of finished board
- Missing Gerber files
- Etc…

Sustainability Impact

IF WE SPECIFY MORE CLEARLY

- Less delays during setup of new parts
- Less risk of manufacturing errors and thus, reduced scrap
- Less labor needed for clarification, increasing profitability

17 Over specification

NCAB Group also experiences cases / examples of over-specification (too much information) and while with under specification the time is spent searching for missing information, here the time and effort is spent of reviewing every single piece of information and determining what parts are critical for the PCB. As may be imagined this can cause delays.

EXAMPLES OF OVER-SPECIFICATION

- Long specification, takes time to understand
- References to specifications not widely used, such as DIN, BS, IEC etc.
- Over specification can lead to contradictory information
- Over specification can lead to excessively high demands / reliability classification despite purpose

Sustainability Impact

IF WE HAVE TOO MUCH OR UNNECESSARY INFORMATION

- Less delays for clarification during setup of new parts
- Less risk of manufacturing errors and thus, reduced scrap
- Less labor needed for clarification, increasing profitability
- Less risk for excessive/unnecessary demands

18 Reliability

NCAB Group does not often experience demands for high reliability (normally IPC class 3) that are not borne out of end product usage demands. It's easy to specify the highest demand without real or full knowledge of the consequences – in both commercial and supply chain terms. The cost increase involved in achieving such high-reliability demands will be extremely high.

TO BE ABLE TO FULFIL ALL DEMANDS IN ACCORDANCE IPC CLASS 3, THE FOLLOWING THREE MAJOR POINTS MUST BE CONTROLLED

- Design: a tight design might drive the cost or even make it impossible to fulfil the demands
- Process control: a 'serious' factory will have very stable and well-controlled processes
- Verification: the verification on a product level for IPC class 3 is extremely time-consuming and should really only be applied for the most demanding of products

Sustainability Impact

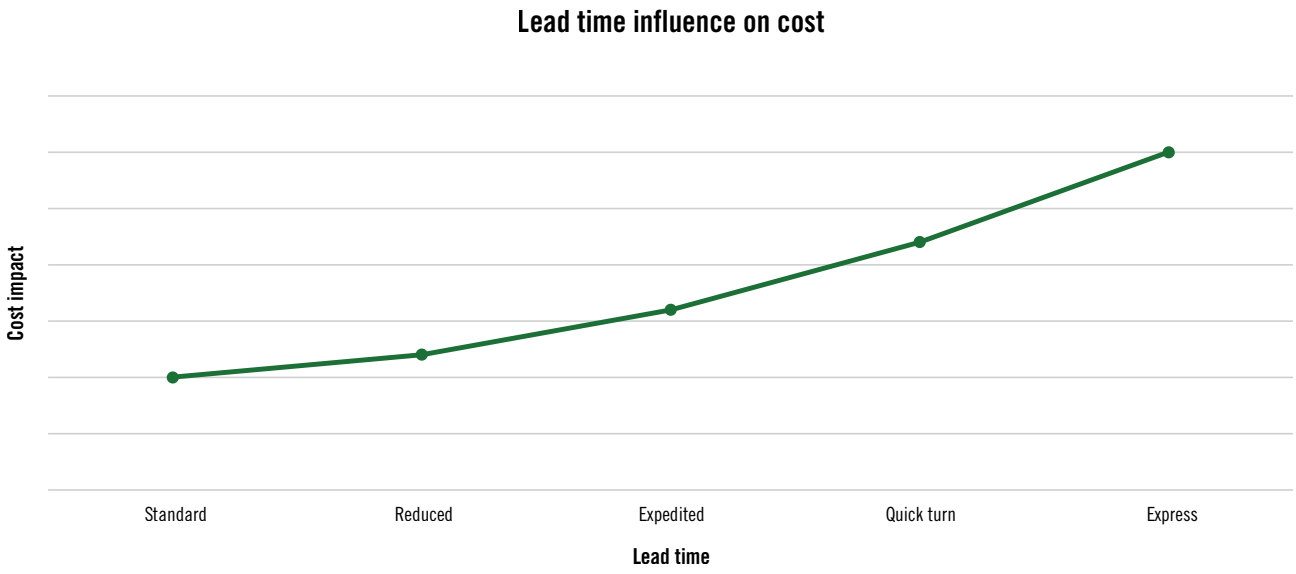
IF WE DESIGN TO IPC CLASS NEEDED RATHER THAN CLASS III WHICH IS NOT NEEDED OFTEN

- Significantly decrease costs/increase profitability
- Increase factory availability and competitive pricing
- Decrease labor needed for verification

19 Lead times

Faster turn times increase cost exponentially. It puts additional strain on the factory to successfully complete the build on a shortened turn time. The graphic shows the average multiplier for each turn time.

20 days	**x1 (Base price)**
15 days	**x1.2**
10 days	**x1.5**
5 days	**x2**
3 days	**x2.5**



Sustainability Impact

IF WE ARE ABLE TO MANAGE WITH LONGER LEAD TIMES

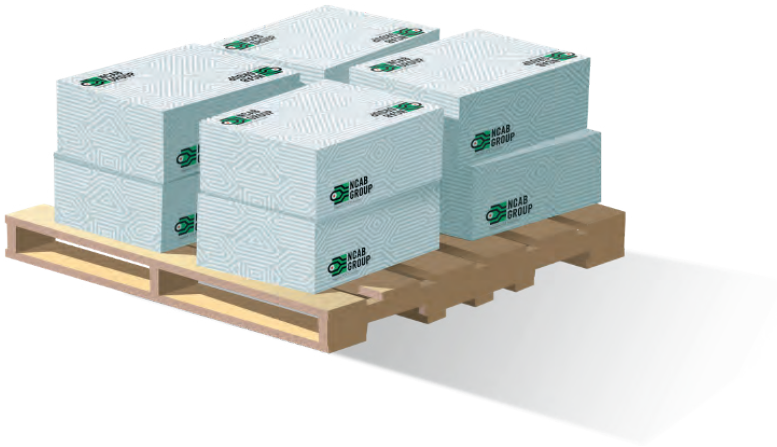
- Increased the number of factories that can compete
- Maximized shipping methods available reducing the carbon footprint

20 Transportation costs

The cost of moving PCBs around the globe can be significant. Multiple options are available such as standard air, express, expedited and sea freight for example. Available options are dependent on location of shipper and receiver and not all options are available for all locations.

Freight is generally a cost vs. transport time trade off cost driver. It is dependent on weight, and weight can be influenced by bare PCB size, thickness, copper weights, etc.

Transportation can be a varied sustainability driver. Expedited freight methods typically have a larger carbon footprint than slower methods.



Sustainability Impact

IF WE ARE ABLE TO CHOOSE VARIOUS FREIGHT METHODS

- Co2 emissions can be reduced with various options
- Cost can be reduced increasing profitability

21 Lack of knowledge, communication, willingness

It is very common that bad design, unsuitable material, wasteful panel layouts, etc. is built in at an early stage due to a lack of knowledge, communication with the volume provider or willingness to challenge 'the current way.'

When faced with critical time-to-market situations, it is all too easy to say: **"it doesn't matter, this is just the prototype, we can look at or fix this later…"**

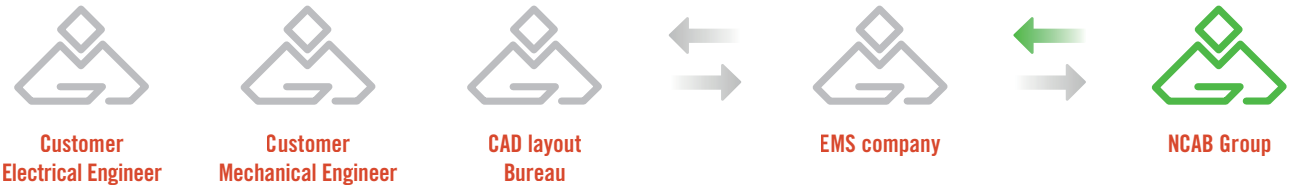
This can lead to delays or problems when the build gets to volume production. NCAB recommends working with our Field Application Engineers, at the earliest design stages, in order to optimize the design.

Sustainability Impact

IF WE ARE ABLE TO COMMUNICATE WITH THE EXPERTS FROM THE START OF THE DESIGN PHASE

- Profitability increases with more knowledge
- More options become available for factory availability
- More options become available for shipping
- We can reduce weight with the most optimal design and materials
- Reduce amount of labor necessary throughout manufacturing
- Overall increase profitability

PCB Cost Drivers
– summary



In the above graphic we see a simplified illustration of a typical PCB supply chain. All of the parties involved in this supply chain share equal responsibility in the success of the specific PCB. This success can only be achieved through open dialog and collaboration from all of the different party members. By far, doing it right and starting collaboration as early as possibly is the best cost saver, and best way to efficiently maximize your ability to create the most sustainable PCB. It's your initiative to seek the support. It's our mission to provide the support.